# RAJALAKSHMI ENGINEERING COLLEGE

**RAJALAKSHMI NAGAR, THANDALAM – 602 105**

## CS23334 Fundamentals of Data Science

## Laboratory Record Note Book

Name : Harine S

Year / Branch / Section : II / CSE / E

University Register No. : 2116250701219

College Roll No. : 250701219

Semester : III

Academic Year : 2026-27

# DEPARTMENT OF COMPUTER SCIENCE

# AND ENGINEERING

## CS23334 Fundamentals of Data Science

## Harine S

| | |
|---|---|
| NAME | HARINE S |
| ROLL NO | 250701219 |
| YEAR | II |
| SEC | E |
| | |

# RAJALAKSHMI ENGINEERING COLLEGE

An Autonomous Institution

# BONAFIDE CERTIFICATE

Name: HARINE S

Academic Year: 2026-27 Semester: III Branch: CSE

Register No. 2116250701219

Certified that this is the bonafide record of work done by the above student in the CS23334 FUNDAMENTALS
OF DATA SCIENCE Laboratory

Signature of Faculty in-charge

Submitted for the Practical Examination held on ……

Internal Examiner External Examiner

| Sl.No | Experiment | Date | GitHub Link |
|---|---|---|---|
| 1.a | Analyze the trend of data science job postings over the last decade for the given data. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP_1.pdf |
| 1.b | Analyze and visualize the distribution of various data science roles. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP_1.pdf |
| 1.c | Differentiate Structured, Un-structured and Semi-structured data. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP_1.pdf |
| 1.d | Encrypt and decrypt given sensitive data using cryptography library fernet. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP_1.pdf |
| 2 | Upload and Analyze the data set given in csv format and perform data preprocessing and visualization. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%202.pdf |
| 3.a | Data visualization using line plot. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%203.pdf |
| 3.b | Data visualization using bar chart. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%203.pdf |
| 3.c | Data visualization using pie chart. | 20-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%203.pdf |
| 4 | Count the frequency of occurrence of a word in a body of text. | 27-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%204.pdf |
| 5 | Data Collection and Initial Exploration. | 27-07-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%205.pdf |
| 6 | Time Series Analysis of Website Traffic. | 03-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%206.pdf |

| Sl.No | Experiment | Date | GitHub Link |
|---|---|---|---|
| 7 | Random Sampling and Sampling Distribution. | 03-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%207.pdf |
| 8 | Conduct Z test for the Data given using Python. | 03-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%208.pdf |
| 9 | Experiment to understand Matplotlib library use cases in Data Science through visualization. | 10-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%209.pdf |
| 10 | Experiment to understand array function in Data science. | 10-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2010.pdf |
| 11 | Experiment to understand pandas library use cases in Data science. | 10-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2011.pdf |
| 12 | Experiment to detect outliers in a given data set. | 17-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2012.pdf |
| 13 | Experiment to handle missing values and inappropriate data in a given data set. | 17-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2013.pdf |
| 14 | Experiment to understand feature scaling. | 17-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2014.pdf |
| 15 | Experiment to understand the data preprocessing in Data science. | 24-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2015.pdf |
| 16 | Experiment to understand EDA - Quantitative and Qualitative analysis. | 24-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2016.pdf |
| 17 | Experiment to understand Linear Regression for a given data set. | 31-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2017.pdf |
| 18 | Experiment to understand KNN algorithm for a given data set. | 31-08-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2018.pdf |

| Sl.No | Experiment | Date | GitHub Link |
|---|---|---|---|
| 19 | Experiment to understand Logistic Regression for a given data set. | 28-09-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2019.pdf |
| 20 | Experiment to understand K-means clustering algorithm for a given data set. | 28-09-2026 | https://github.com/harinesk/FDS-Assignment/blob/main/EXP%2020.pdf |